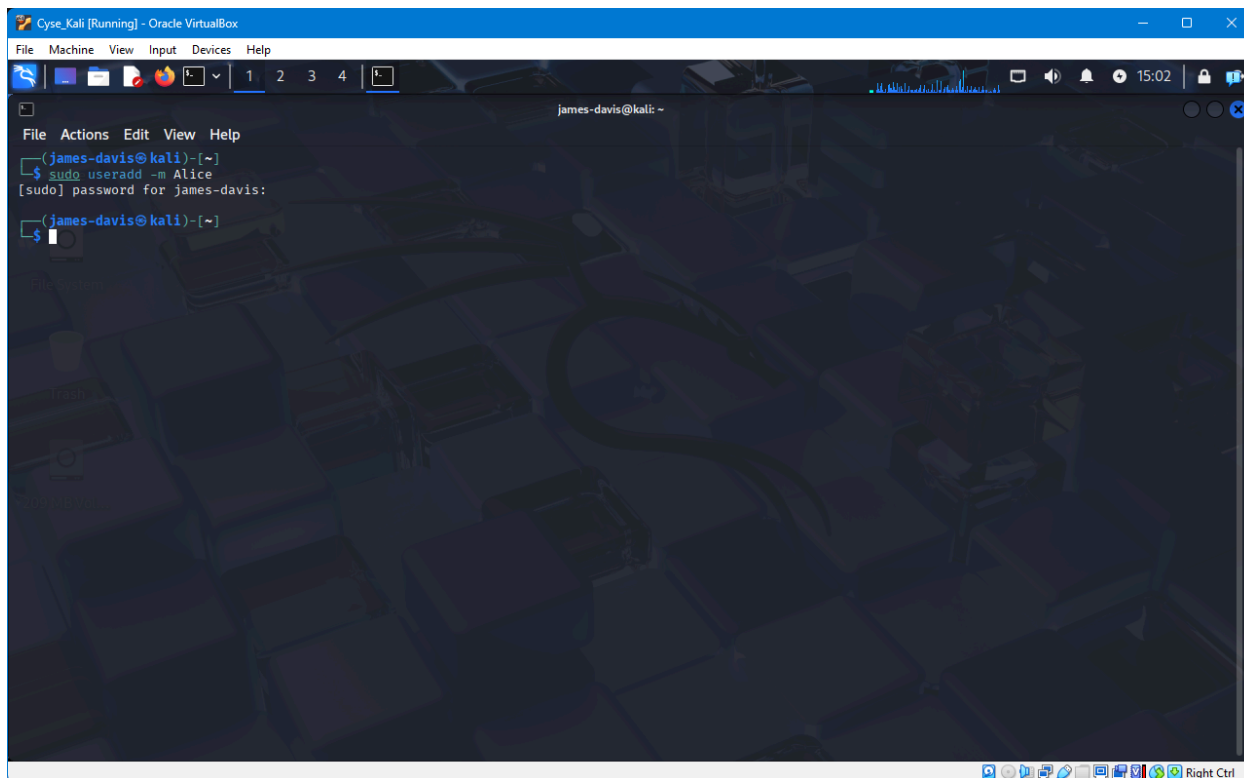


CYSE 270 - Assignment 9: Task Automation - James Davis

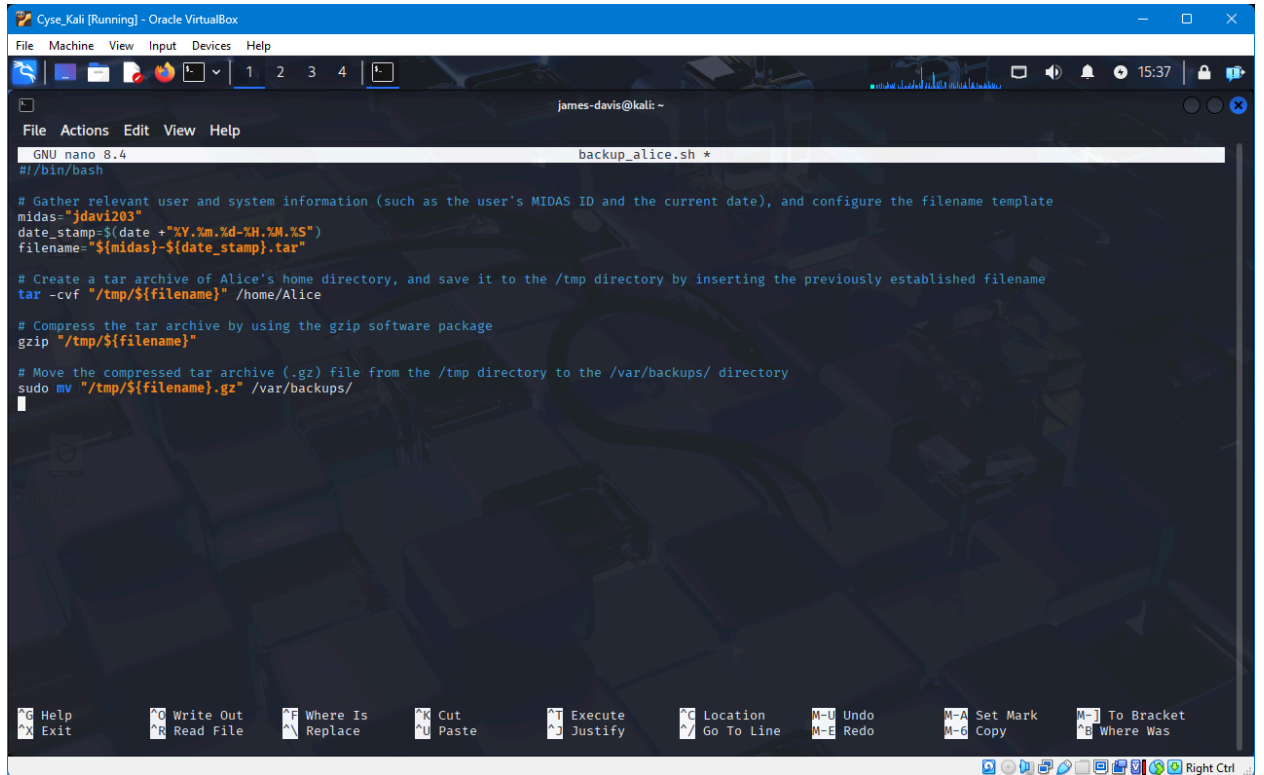
Task A - Backup your system (Using “cronab”)

Step 1: Create a new user named “Alice” (with home directory)



Step 2: Write a shell script that backups Alice's home directory by creating a tar file (tape archive), by satisfying the following requirements:

- Obtains the following user and system information:
 - The user's MIDAS ID.
 - The current date.
- Creates a variable named "filename," consisting of the following content:
 - "[MIDAS ID]-date"
- Creates a tape archive (using the "tar" command) for Alice's home directory (also known as "/home/Alice"), and saves it to the following directory each time:
 - "/tmp/\${filename}"
- Compresses the tape archive using "gzip."
- Moves the compressed tape archive to the "/var/backups" directory.



```
Cyse_Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 | L
james-davis@kali: ~
File Actions Edit View Help
GNU nano 8.4 backup_alice.sh *
#!/bin/bash

# Gather relevant user and system information (such as the user's MIDAS ID and the current date), and configure the filename template
midas="jdavi203"
date_stamp=$(date +"%Y.%m.%d-%H.%M.%S")
filename="${midas}-${date_stamp}.tar"

# Create a tar archive of Alice's home directory, and save it to the /tmp directory by inserting the previously established filename
tar -cvf "/tmp/${filename}" /home/Alice

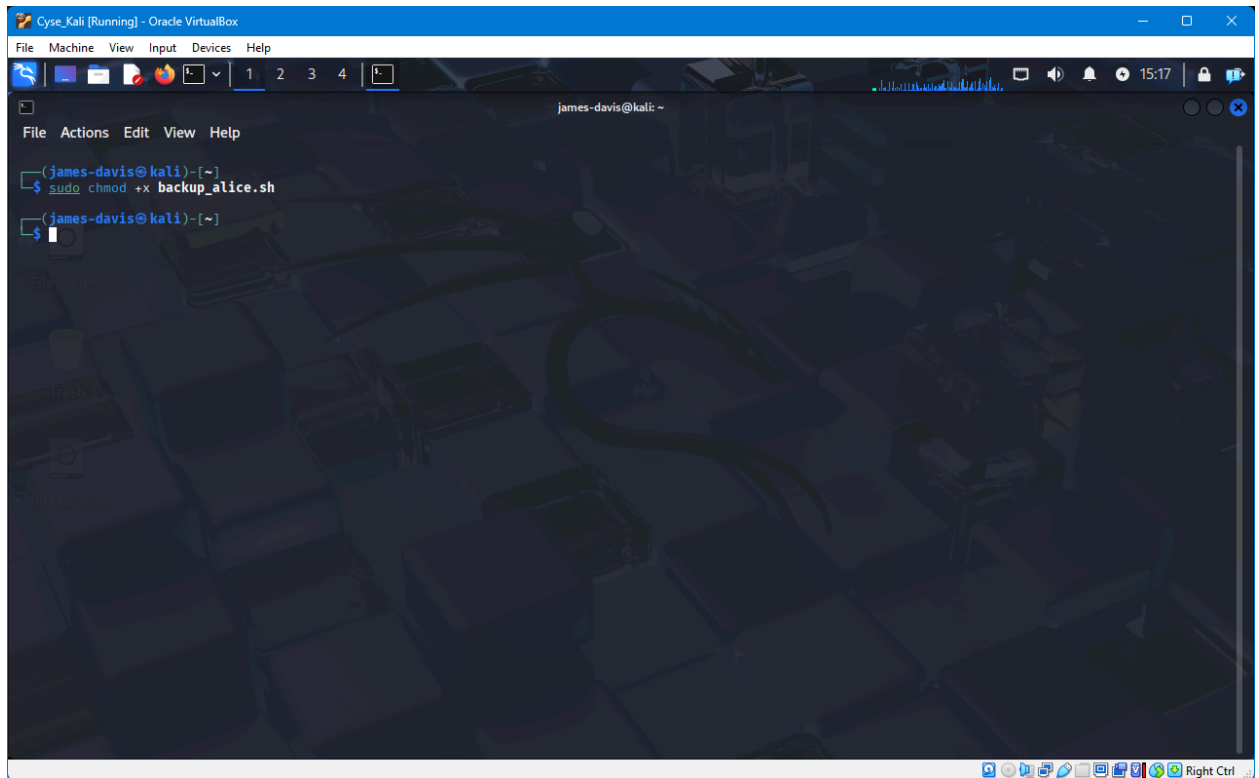
# Compress the tar archive by using the gzip software package
gzip "/tmp/${filename}"

# Move the compressed tar archive (.gz) file from the /tmp directory to the /var/backups/ directory
sudo mv "/tmp/${filename}.gz" /var/backups/

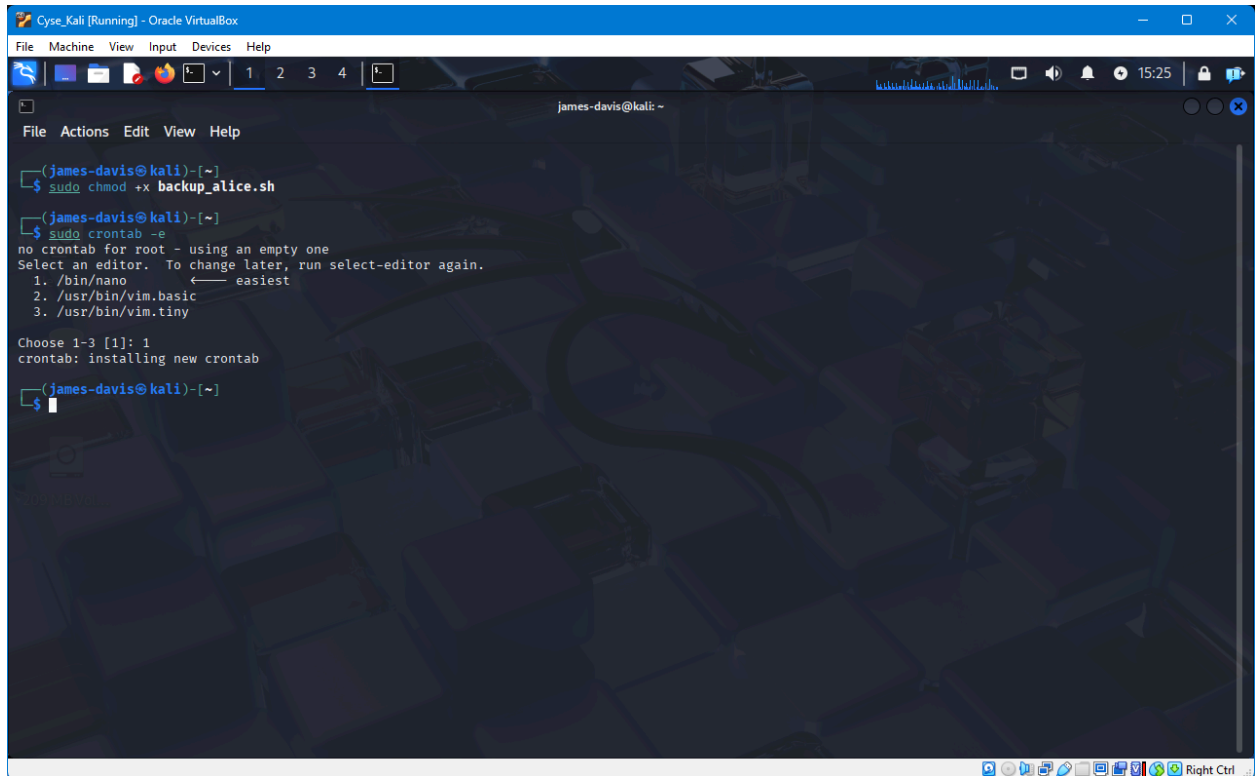
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was

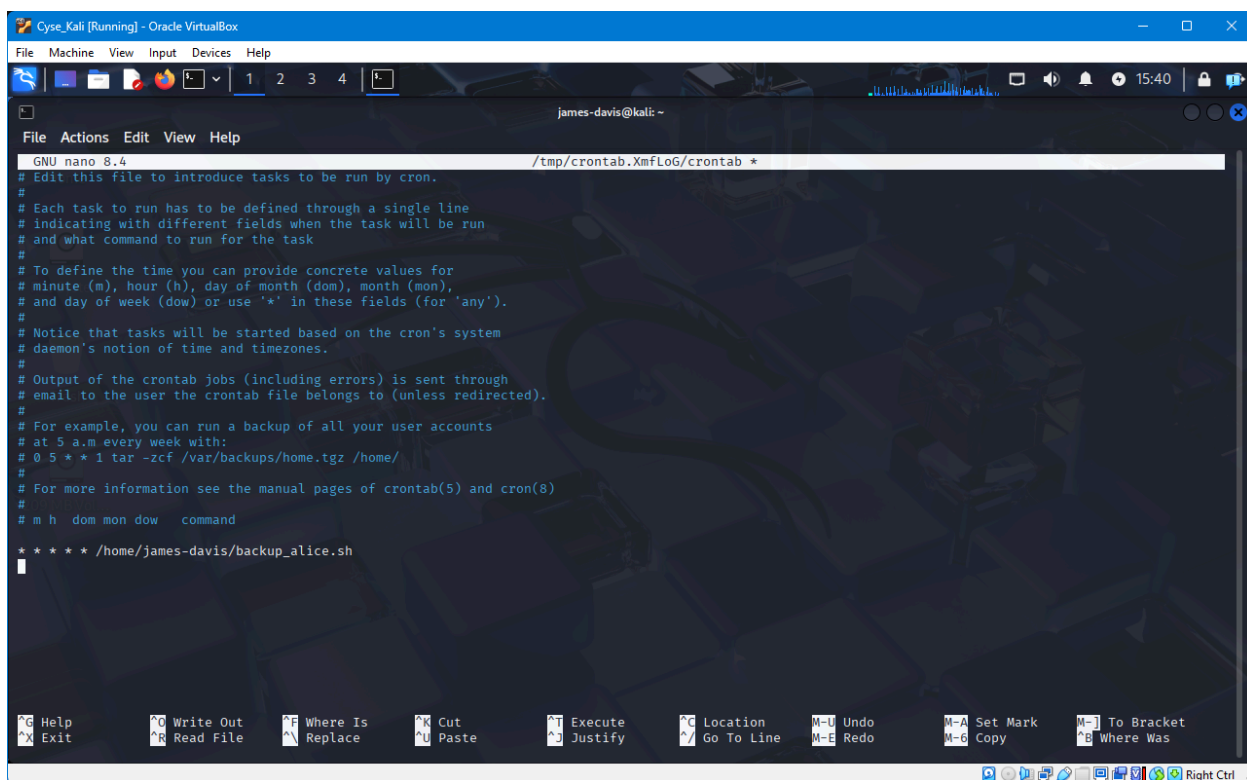
Step 3: Make the “backup_alice.sh” script executable



Step 4: Create a crontab file (which is owned by root, so “sudo” privileges are needed) to ensure the scheduled task runs for 3 minutes (resulting in at least 3 backup file iterations, because the smallest time unit in crontab is “minutes”)

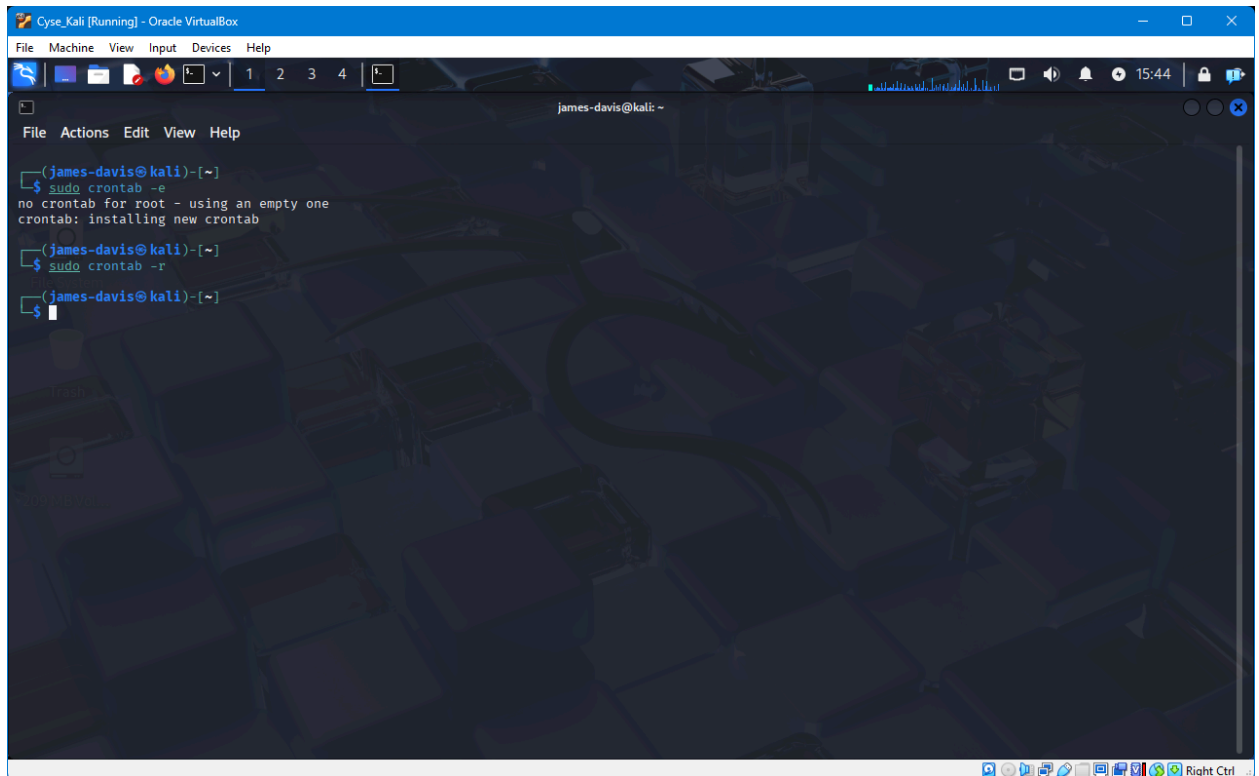


```
Cyse_Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 [5]
james-davis@kali: ~
File Actions Edit View Help
(james-davis@kali)-[~]
$ sudo chmod +x backup_alice.sh
(james-davis@kali)-[~]
$ sudo crontab -e
no crontab for root - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
Choose 1-3 [1]: 1
crontab: installing new crontab
(james-davis@kali)-[~]
$
```



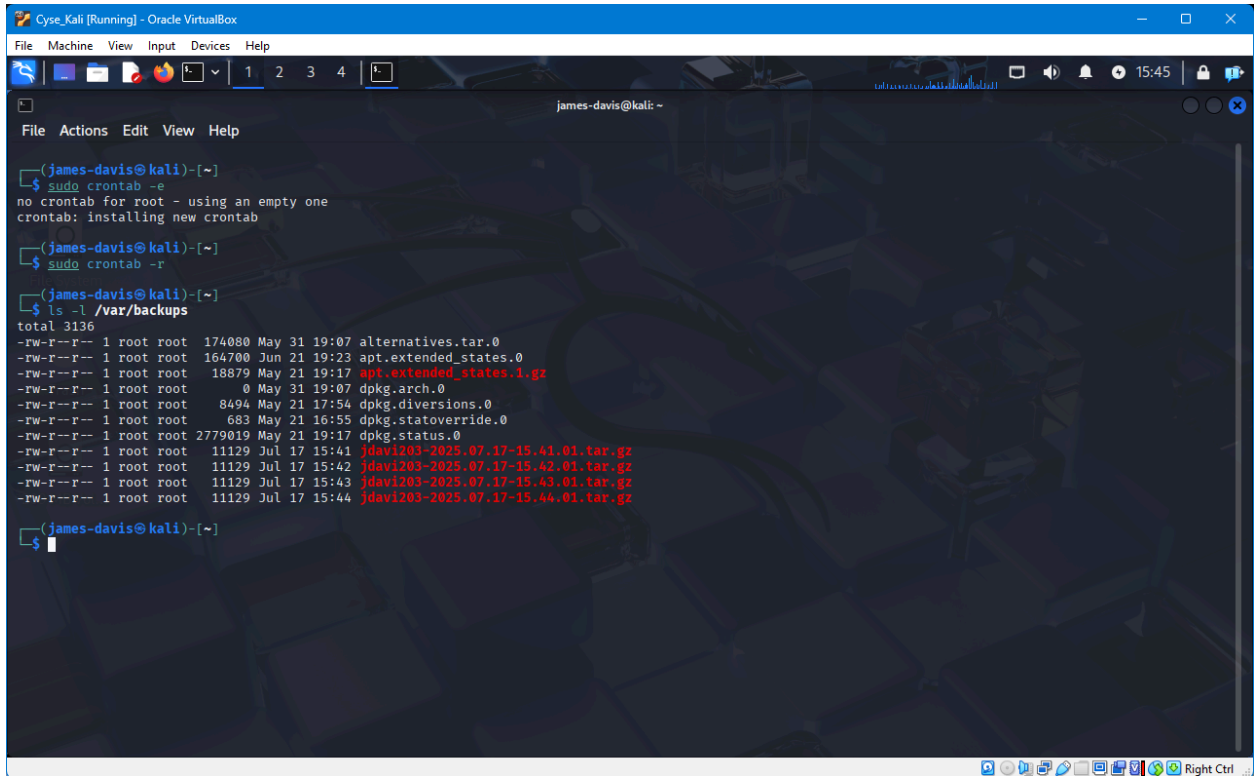
```
Cyse_Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 | 5
james-davis@kali: ~
File Actions Edit View Help
GNU nano 8.4 /tmp/crontab.XmfLoG/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m. every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
* * * * * /home/james-davis/backup_alice.sh
|
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-G Copy      ^B Where Was
```

Step 5: Cancel the crontab jobs



```
Cyse_Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 [5]
james-davis@kali: ~
File Actions Edit View Help
(james-davis@kali)-[~]
$ sudo crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
(james-davis@kali)-[~]
$ sudo crontab -r
(james-davis@kali)-[~]
$
```

Step 6: Check the contents of the “/var/backups” directory



The screenshot shows a Kali Linux terminal window titled "Cyse_Kali [Running] - Oracle VirtualBox". The terminal displays the following commands and output:

```
(james-davis@kali)-[~]
$ sudo crontab -e
no crontab for root - using an empty one
crontab: installing new crontab

(james-davis@kali)-[~]
$ sudo crontab -r

(james-davis@kali)-[~]
$ ls -l /var/backups
total 2136
-rw-r--r-- 1 root root 174080 May 31 19:07 alternatives.tar.0
-rw-r--r-- 1 root root 164700 Jun 21 19:23 apt.extended_states.0
-rw-r--r-- 1 root root 18879 May 21 19:17 apt.extended_states.1.gz
-rw-r--r-- 1 root root 0 May 31 19:07 dpkg.arch.0
-rw-r--r-- 1 root root 8494 May 21 17:54 dpkg.diversions.0
-rw-r--r-- 1 root root 683 May 21 16:55 dpkg.statoverride.0
-rw-r--r-- 1 root root 2779019 May 21 19:17 dpkg.status.0
-rw-r--r-- 1 root root 11129 Jul 17 15:41 jdavi203-2025.07.17-15.41.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:42 jdavi203-2025.07.17-15.42.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:43 jdavi203-2025.07.17-15.43.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:44 jdavi203-2025.07.17-15.44.01.tar.gz

(james-davis@kali)-[~]
$
```

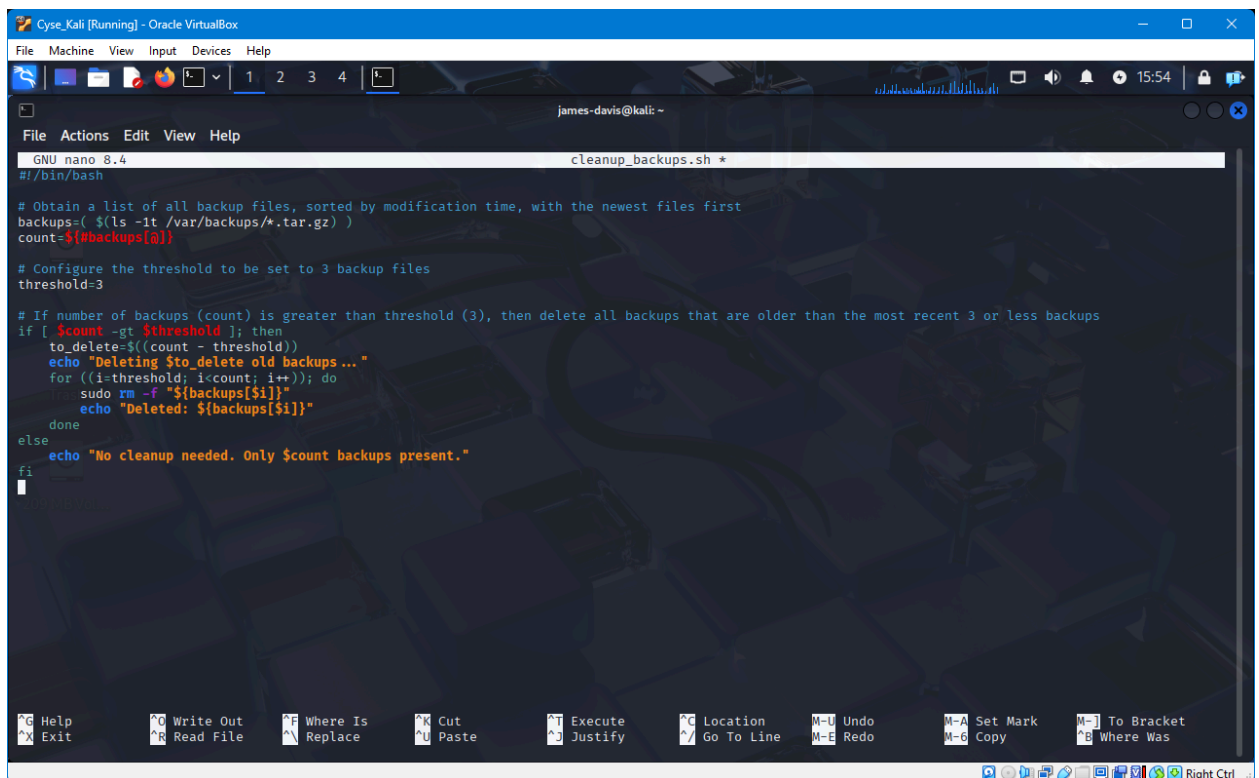
The terminal window includes a menu bar (File, Machine, View, Input, Devices, Help) and a toolbar with icons for file operations. The background of the terminal is a dark, abstract pattern. The bottom of the window shows a taskbar with various application icons and a system tray with a clock and network status.

Task B - System Cleanup (Extra Credit)

Step 1: Write a script that cleans up the old backups, which checks the number of backups that were created in Task A, and determines whether to delete archives that are deemed outdated/unnecessary by maintaining backups under a pre-defined threshold, by meeting the following requirements:

- Counts the number of backups from Task A, and:
 - Determines if this number is greater than 3.
 - If the number is less than 3 (or equal to 3), then do nothing (because no backups would be too outdated/unnecessary to keep yet).
 - If the number is greater than 3, then:
 - Calculate the number of backups to delete.
 - Delete the old archives.
 - Achieve this by having the “backups” variable:
 - Interpret a list of one file per line.
 - Sort the files by modification time, with the newest files interpreted first, meaning:
 - Newest kept = 0 (the first file).
 - Oldest kept = 2 (the third file).
 - And by having the “for” loop:

- Delete the backups that contain a current ID number of 3 or greater (the fourth file or greater)
 - Because this script needs to be able to delete old backups in the “/var/backups” directory (owned by root), “sudo” privileges are needed.



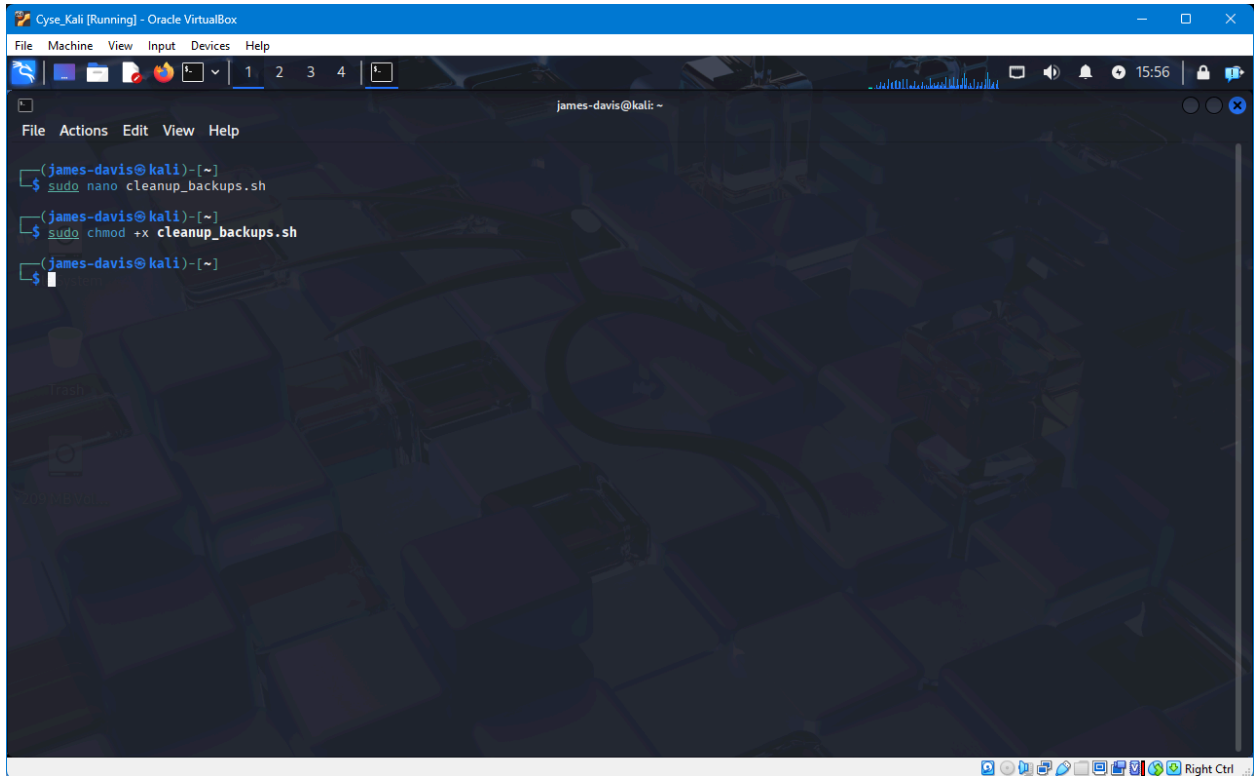
```
Cyse_Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 |
james-davis@kali: ~
File Actions Edit View Help
GNU nano 8.4 cleanup_backups.sh *
#!/bin/bash

# Obtain a list of all backup files, sorted by modification time, with the newest files first
backups=( $(ls -lt /var/backups/*.tar.gz) )
count=${#backups[@]}

# Configure the threshold to be set to 3 backup files
threshold=3

# If number of backups (count) is greater than threshold (3), then delete all backups that are older than the most recent 3 or less backups
if [ $count -gt $threshold ]; then
    to_delete=$((count - threshold))
    echo "Deleting $to_delete old backups..."
    for ((i=threshold; i<count; i++)); do
        sudo rm -f "${backups[$i]}"
        echo "Deleted: ${backups[$i]}"
    done
else
    echo "No cleanup needed. Only $count backups present."
fi
```

Step 2: Make the “cleanup_backups.sh” script executable



The screenshot shows a terminal window titled "Cyse_Kali [Running] - Oracle VM VirtualBox". The terminal prompt is "james-davis@kali: ~". The user has executed the following commands:

```
(james-davis@kali)-[~]  
$ sudo nano cleanup_backups.sh  
(james-davis@kali)-[~]  
$ sudo chmod +x cleanup_backups.sh  
(james-davis@kali)-[~]  
$
```

The terminal window has a menu bar with "File", "Actions", "Edit", "View", and "Help". The background of the terminal is a dark blue isometric cityscape. The window title bar includes standard window controls and a system tray with icons for network, volume, and power, along with the time "15:56".

Step 3: Test the “cleanup_backups.sh” script to ensure that it accurately reports how many backup files were deleted, and that the oldest backup file(s) was/were deleted. Furthermore, verify that “/var/backups” directory now only contains the most recent 3 backup files, instead of 4 or more backup files

```

Cyse_Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4

james-davis@kali: ~
File Actions Edit View Help
$ sudo chmod +x cleanup_backups.sh
(james-davis@kali)-[~]
$ ls -l /var/backups
total 3136
-rw-r--r-- 1 root root 174080 May 31 19:07 alternatives.tar.0
-rw-r--r-- 1 root root 164700 Jun 21 19:23 apt.extended_states.0
-rw-r--r-- 1 root root 18879 May 21 19:17 apt.extended_states.1.gz
-rw-r--r-- 1 root root 0 May 31 19:07 dpkg.arch.0
-rw-r--r-- 1 root root 8494 May 21 17:54 dpkg.diversions.0
-rw-r--r-- 1 root root 683 May 21 16:55 dpkg.statoverride.0
-rw-r--r-- 1 root root 2779019 May 21 19:17 dpkg.status.0
-rw-r--r-- 1 root root 11129 Jul 17 15:41 jdavi203-2025.07.17-15.41.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:42 jdavi203-2025.07.17-15.42.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:43 jdavi203-2025.07.17-15.43.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:44 jdavi203-2025.07.17-15.44.01.tar.gz

(james-davis@kali)-[~]
$ ./cleanup_backups.sh
Deleting 1 old backups ...
Deleted: /var/backups/jdavi203-2025.07.17-15.41.01.tar.gz

(james-davis@kali)-[~]
$ ls -l /var/backups
total 3124
-rw-r--r-- 1 root root 174080 May 31 19:07 alternatives.tar.0
-rw-r--r-- 1 root root 164700 Jun 21 19:23 apt.extended_states.0
-rw-r--r-- 1 root root 18879 May 21 19:17 apt.extended_states.1.gz
-rw-r--r-- 1 root root 0 May 31 19:07 dpkg.arch.0
-rw-r--r-- 1 root root 8494 May 21 17:54 dpkg.diversions.0
-rw-r--r-- 1 root root 683 May 21 16:55 dpkg.statoverride.0
-rw-r--r-- 1 root root 2779019 May 21 19:17 dpkg.status.0
-rw-r--r-- 1 root root 11129 Jul 17 15:42 jdavi203-2025.07.17-15.42.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:43 jdavi203-2025.07.17-15.43.01.tar.gz
-rw-r--r-- 1 root root 11129 Jul 17 15:44 jdavi203-2025.07.17-15.44.01.tar.gz

```